Comment on “Multiobjective model predictive control” [Automatica 45 (2009) 2823–2830]

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Abstract

Lemma 4 of the commented paper states that the value function of the multi-parametric linear program with weight vector μ in the cost and state x in the right-hand side is convex and piecewise affine in μ for fixed x. The function is piecewise affine, but on its domain it is *concave* in μ, not convex: for each fixed decision variable the objective is affine in μ and the feasible set does not depend on μ, so the value function is a pointwise minimum of affine functions. We trace the slip to a duality remark in the companion conference paper, and we work out the consequences. The weight-selection linear program of Theorem 6 evaluates the value function as the *maximum* of its affine pieces and therefore imposes all of them at once; it describes a convex inner approximation of the true admissible weight set, which is in general non-convex. Every weight it returns satisfies the contractive constraint, so the closed-loop guarantees that rest on that constraint are preserved; but the claimed equivalence does not hold, and on randomly generated instances of the commented problem the program is strictly conservative in 97.5% of the levels tested and infeasible while admissible weights exist in 62.9% of them. In the quadratic case, the joint convexity invoked in Theorem 7 does not hold, because the bilinear coupling between weights and decision variables makes the joint Hessian indefinite unless

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that coupling vanishes; the online problem is not a convex program. The first defect admits an exact repair: since the value function is the minimum of the affine pieces, the admissible weight set is their *union* of half-spaces, obtained exactly by solving one linear program per active piece — each smaller than the published one and independent of the others, their number growing with the number of active pieces — which also restores the recursive feasibility witness that the published formulation loses. For the quadratic case we show that any feasible point of the published program still satisfies the contractive constraint, so only global optimality is at stake.

Keywords: Predictive control, multiobjective optimization, multiparametric programming

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1. The claim

The commented paper [1] builds a multiobjective MPC scheme in which, at each sample, a weight vector is selected online subject to the contractive constraint $V^*(\mu, x) \leq J_a$, where V^* is the optimal value of a multiparametric program with the weights in the cost. For the piecewise affine case, the machinery rests on [1, Lemma 4], which considers the multiparametric linear problem—their problem (13)—with parameters $\mu \in \mathbb{R}^l$ in the cost function and $x \in \mathbb{R}^n$ in the right-hand side of the constraints, and states that the value function V^* is

> "continuous w.r.t. (μ, x), convex and piecewise affine w.r.t. μ for any given x and w.r.t. x for any given μ."

The statement for x is correct, and so is piecewise affinity in μ. The convexity claim for μ, however, has the wrong sign.

2. The correction

Lemma 1. *Let $V^*(\mu, x) = \min_z\{(c_0 + C'_\mu \mu)' z : \; Gz \leq b + Sx\}$ be the value function of problem (13) of [1]. Fix x such that the feasible set is nonempty, and* *let $D_x = \{\mu : V^*(\mu, x) > -\infty\}$ be the set of weights for which the program is bounded. Then $\mu \mapsto V^*(\mu, x)$ is concave on D_x, and piecewise affine there whenever the feasible set is pointed — as it is in the setting of the commented paper, whose decision vector consists of the input sequence, confined to the bounded set U [1, footnote 3], and of non-negative slack variables, so that all its components are bounded below (see the sentence preceding their Eq. (14)).*

Proof. For each fixed feasible z the map $\mu \mapsto (c_0 + C'_\mu \mu)' z$ is affine, and the feasible set $\{z : Gz \leq b + Sx\}$ does not depend on μ. Hence $V^*(\cdot, x)$ is a pointwise infimum of a family of affine functions of μ, and therefore concave wherever it is finite. If the feasible set is pointed, the infimum is attained at one of its finitely many vertices, so on D_x the function is the minimum of finitely many affine functions and hence piecewise affine. □

This is the classical sensitivity dichotomy of parametric optimization: under minimization the optimal value is convex in right-hand-side perturbations and concave in parameters that enter the objective linearly [3, Sec. 5.6.1, Ex. 3.9]. Both signs invert under maximization. The origin of the slip appears to be the companion conference paper [2, p. 2406], which establishes the properties in x and adds that "by duality, the same properties hold with respect to μ." Duality does exchange right-hand-side and cost parameters, but it also turns the primal minimum into a dual maximum, and that is precisely the step that flips the curvature.

Numerical verification on the commented problem. We verified the corrected statement directly on problem (13), on randomly generated instances rather than on any particular controller, so that the evidence does not depend on how the scheme is instantiated. 200 random multiparametric linear programs with six decision variables, eight constraints and two weights, the weights sampled in the simplex $\{\mu \geq 0,\ \sum_i \mu_i \leq 1\}$ declared in the commented paper, and six chord tests each (1,200 tests): the concavity inequality in μ held in every test, strictly in 746 of them and with equality in the rest, where the chord lay inside a single affine piece; the convexity inequality in μ held strictly in none. The same test in x — the direction the commented paper gets right — came out convex in every test, strictly in 295. A third test confirms the direction of the error at the level of the pieces: evaluating the value function as the *maximum* of the affine pieces, as the proof of Theorem 6 does, overstates it by up to 1.34 in value over the simplex, while the *minimum* reproduces it to 4×10^{-15}. Code and data are available with this comment.

3. Consequences within the commented paper

Theorem 6 and the weight-selection LP (17). The proof of Theorem 6 in [1] invokes the result of Schechter [5] for *convex* piecewise affine functions to evaluate $V^*(\mu, x)$, for fixed x, as the *maximum* of the affine pieces $\{(\phi_i x + \gamma_i)'(c_0' + C_\mu' \mu)\}_{i \in I(x)}$, and the linear program (17) accordingly imposes

$$(\phi_i x + \gamma_i)'(c_0' + C_\mu' \mu) \;\leq\; J_a \qquad \text{for all } i \in I(x).$$

By Lemma 1 the value function is the *minimum* of those pieces. Requiring every piece to lie below J_a is therefore sufficient but not necessary for $V^*(\mu, x) \leq J_a$: the constraint set of (17) is a convex *inner approximation* of the true admissible weight set

$$\mathcal{A}(x, J_a) = \{\mu : \; V^*(\mu, x) \leq J_a\},$$

which, being a sublevel set of a concave function, is in general non-convex (for scalar μ it is a union of at most two intervals whenever the maximum of $V^*(\cdot, x)$ is interior and exceeds J_a).

Two things follow, one reassuring and one not. Since every weight accepted by (17) satisfies the contractive constraint, the closed-loop guarantees that rest on that constraint are untouched, and the piecewise affine scheme is safe as published. What fails is exactness. We quantified the gap on 480 instance–level pairs: 120 random instances, each at four contraction levels taken as the 15th, 35th, 55th and 75th percentiles of $V^*(\cdot, x)$ over a grid of the weight simplex $\{\mu \geq 0, \; \sum_i \mu_i \leq 1\}$ declared in the commented paper, so that the admissible set is never empty nor the whole simplex. (17) was always contained in $\mathcal{A}(x, J_a)$, as the theory predicts, but *strictly* contained in 468 of them (97.5%); $\mathcal{A}(x, J_a)$ was non-convex in 232 (48.3%); and, most consequentially, (17) was *infeasible while admissible weights existed* in 302 of them (62.9%); that rate falls with the level, from 79.2% at the lowest to 45.0% at the highest, but never vanishes. In those samples the published program reports that no weight meets the contraction level when in fact a set of them does. Its optimizer, when feasible, also need not be the admissible weight closest to the desired one.

Theorem 7 and the quadratic case. For the branch with one quadratic objective, the proof of Theorem 7 in [1] asserts joint convexity of $V^*_\mu(\mu, x)$ in (μ, x) "by the results of Mangasarian and Rosen (1964)," and concludes that the online problem (21) is a convex program. In their problem (20) the weights enter the objective linearly through the bilinear term $\mu' C_\mu z$ while the constraints $Gz \leq b + Sx$ do not depend on μ, so the Hessian of the objective with respect to (z, μ) has the block form

$$\begin{bmatrix} H & C'_\mu \\ C_\mu & 0 \end{bmatrix},$$

which is positive semidefinite only if $C_\mu = 0$, that is, only if the weights do not act at all. The objective is therefore not jointly convex in (z, μ), and partial minimization over z cannot be invoked to conclude joint convexity of the value function. Independently, the argument of Lemma 1 applies verbatim in the μ direction, so $V^*_\mu(\cdot, x)$ is concave; a function that is both concave and convex in μ is affine in μ, so joint convexity can hold only in that degenerate case. Consequently the feasible set of (21) is in general non-convex and (21) is not a convex program. The closed-loop guarantee is preserved here too: the contractive constraint is imposed exactly in (21), so any feasible point satisfies it, and a tangent linearization of a concave constraint is conservative. What is lost is global optimality — a local solver may stop at a suboptimal weight, or report infeasibility where admissible weights exist — and warm-starting at μ_{t-1}, the scheme's own recursive-feasibility witness, removes the spurious infeasibility.

4. An exact repair for the piecewise affine case

The correction also supplies the fix. Because $V^*(\cdot, x)$ is the *minimum* of the pieces, the admissible set is their *union* of half-spaces rather than their intersection:

$$\mathcal{A}(x, J_a) \;=\; \bigcup_{i \in I(x)} \big\{\mu \in \Delta :\; (\phi_i x + \gamma_i)'(c_0' + C_\mu' \mu) \le J_a\big\}, \tag{1}$$

with Δ the weight simplex. Each member of the union is defined by a *single* affine inequality over the same simplex, so selecting the admissible weight closest to a desired one is solved exactly by $\operatorname{card}(I(x))$ independent linear programs, each with a single contraction row in place of the $\operatorname{card}(I(x))$ rows of (17), and the best of their optima is returned; the price of exactness is that their number grows with the number of active pieces. On the 480 instance–level pairs above, (1) reproduced the true admissible set exactly in 466; the 14 disagreements are grid points lying on the level itself ($V^* = J_a$ within tolerance), and the minimum of the pieces matches the value function to 2×10^{-9}.

The repair also restores a property the published program loses. The recursive feasibility argument of [1, Theorem 3] exhibits the previous weight as a witness: μ_{t-1} satisfies the contraction level by construction, hence $\mathcal{A}(x_t, J_a) \neq \emptyset$. That witness certifies the *minimum* of the pieces, which is exactly what (1) tests, whereas (17) tests their maximum — which is why (17) can be infeasible at a state where the scheme's own witness is available. An implementation that keeps the previous weight whenever (17) is infeasible preserves the contractive constraint and therefore the guarantee; but with (1) the fallback is not needed, because the feasible set is never empty when the witness exists.

Gridding the simplex and testing $V^*(\mu, x) \le J_a$ pointwise remains a practical alternative for very low weight dimensions, but it is an approximation, not an exact method, and we do not recommend it where (1) applies. Projection or gradient steps on the weight, and any construction that presumes convexity of $\mathcal{A}(x, J_a)$, can silently leave the admissible set.

Closing remark

The contribution of the commented paper—a multiobjective MPC formulation with closed-loop guarantees obtained through a contractive constraint on a weighted value function—is not in question, and the published piecewise affine algorithm remains safe precisely because the affected construction errs on the conservative side. The correction concerns the exactness claims attached to the weight-selection subproblems, and for the piecewise affine case it comes with an exact repair whose cost is $\mathrm{card}(I(x))$ small, independent linear programs. The authors of the commented paper were contacted about this observation on 7 August 2026; no response had been received at the time of submission.

Data availability

The scripts and result files that generate every number reported here are distributed with this comment. They were written with the assistance of Claude (Anthropic) from the sampling protocol and acceptance criteria specified by the authors, who ran them and are responsible for their outputs.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic; Claude Opus 4.6, Opus 5 and Fable 5.1), accessed through the Claude Code command-line assistant, in order to draft and revise the prose of this comment, to write the scripts that generate its numerical evidence, and to run adversarial reviews of its argument. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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